

SIMATS ENGINEERING

ASSESSMENT TOOL I — Uncertainty Visualization Lab

Course: Data Handling and Visualization	Code: DSA06	CO: C05
Duration: 180 min	Total Marks: 100	Reg No : 192424102 Name:Yugitha B

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool : Uncertainty Visualization Lab

Weightage: 20%

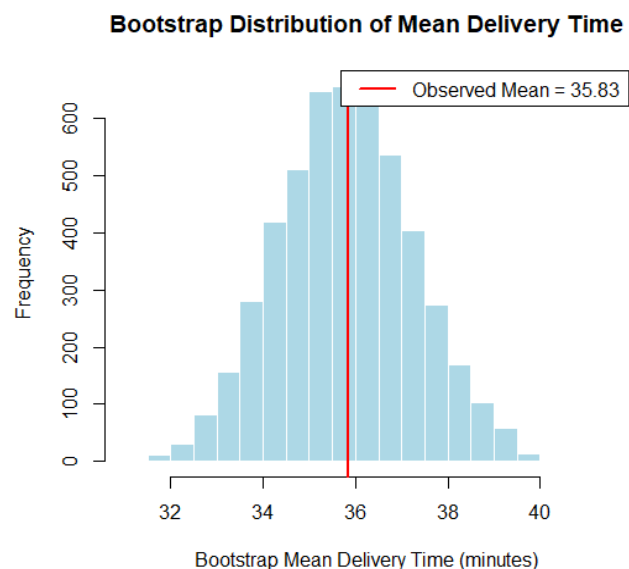
Code of Conduct:

*I **Yugitha B (192424102)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

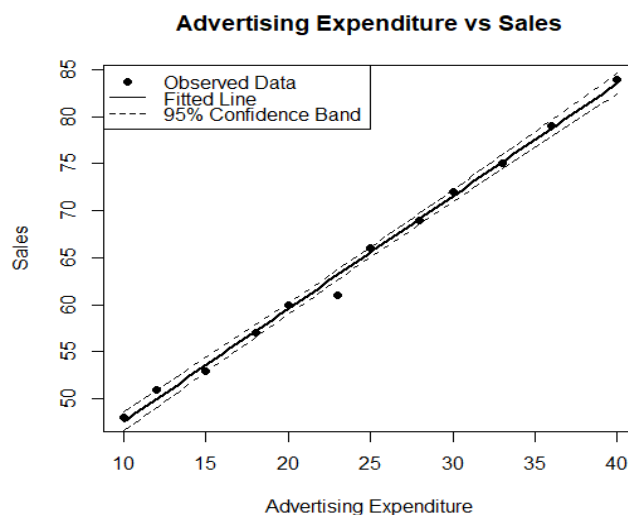
Q1. Bootstrapping and Confidence Intervals – Point Estimate

```
delivery <- c(31, 35, 28, 42, 37, 33, 45, 29, 39, 34, 41, 36)
observed_mean <- mean(delivery)
observed_mean
set.seed(123)
B <- 5000
bootstrap_means <- replicate(B, {
  sample_data <- sample(delivery, size = length(delivery), replace = TRUE)
  mean(sample_data)
})
CI <- quantile(bootstrap_means, c(0.025, 0.975))
CI
hist(bootstrap_means,
     breaks = 30,
     xlab = "Bootstrap Mean Delivery Time (minutes)",
     col = "lightblue",
     border = "white")
abline(v = observed_mean,
       col = "red",
       lwd = 2)
legend("topright",
      legend = paste("Observed Mean =", round(observed_mean, 2)),
      col = "red",
      lwd = 2)
```



Q2. Uncertainty in Trend Estimation

```
Advertising <- c(10, 12, 15, 18, 20, 23, 25, 28, 30, 33, 36, 40)
Sales <- c(48, 51, 53, 57, 60, 61, 66, 69, 72, 75, 79, 84)
data <- data.frame(Advertising, Sales)
model <- lm(Sales ~ Advertising, data = data)
summary(model)
new_data <- data.frame(
  Advertising = seq(min(Advertising), max(Advertising), length.out = 100)
)
pred <- predict(model, newdata = new_data, interval = "confidence")
plot_data <- cbind(new_data, pred)
plot(data$Advertising, data$Sales,
      main = "Advertising Expenditure vs Sales",
      xlab = "Advertising Expenditure",
      ylab = "Sales",
      pch = 19)
lines(plot_data$Advertising, plot_data$lwr,
      lty = 2)
lines(plot_data$Advertising, plot_data$upr,
      lty = 2)
legend("topleft",
      legend = c("Observed Data", "Fitted Line", "95% Confidence Band"),
      pch = c(19, NA, NA),
      lty = c(NA, 1, 2))
```



Q3. Overplotting, Jittering and Transparency

```
library(ggplot2)
```

```
Hours <- c(2,3,3,4,4,4,5,5,5,5,6,6,6,6,  
          7,7,7,7,7,8,8,8,8,8,9,9,9,10,10)
```

```
Scores <- c(45,52,52,58,60,60,64,66,66,66,70,72,72,72,72,  
           76,78,78,80,80,84,84,85,85,85,88,90,90,94,94)
```

```
data <- data.frame(Hours, Scores)
```

```
ggplot(data, aes(x = Hours, y = Scores)) +
```

```
  geom_point() +
```

```
  labs(  
    title = "Original Scatterplot",  
    x = "Hours Studied",  
    y = "Examination Score"  
  ) +
```

```
  labs(  
    title = "Scatterplot with Transparency",  
    x = "Hours Studied",  
    y = "Examination Score"  
  ) +
```

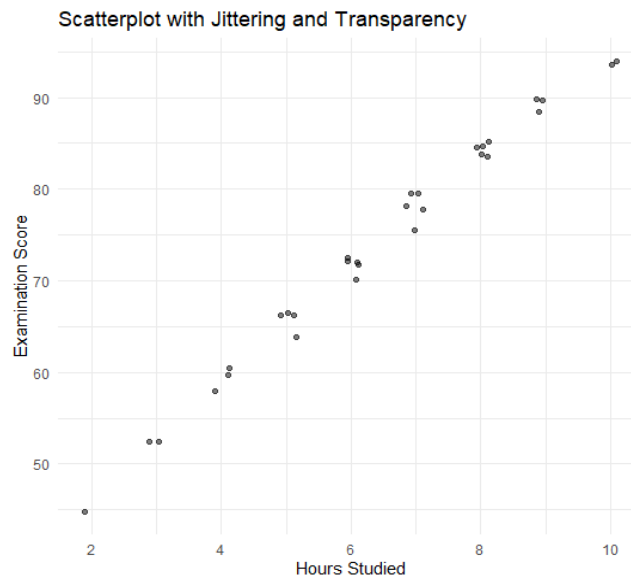
```
  theme_minimal()
```

```
ggplot(data, aes(x = Hours, y = Scores)) +
```

```
  geom_point(  
    alpha = 0.5,  
    position = position_jitter(width = 0.15, height = 0.5)  
  ) +
```

```
  labs(  
    title = "Scatterplot with Jittering and Transparency",  
    x = "Hours Studied",  
    y = "Examination Score"  
  ) +
```

```
theme_minimal()
```



Q4. 2D Histogram and Density Estimation

```
library(ggplot2)
```

```
Height <- c(150,152,155,158,160,161,163,165,166,168,  
            170,171,173,175,176,178,180,182,184,186)
```

```
Weight <- c(48,50,53,55,57,59,60,62,64,65,  
            68,69,71,73,74,77,80,82,85,88)
```

```
data <- data.frame(Height, Weight)
```

```
ggplot(data, aes(x = Height, y = Weight)) +
```

```
  geom_bin2d(bins = 6) +
```

```
  labs(
```

```
    title = "2D Histogram of Height and Weight",
```

```
    x = "Height (cm)",
```

```
    y = "Weight (kg)"
```

```
  ) +
```

```
  theme_minimal()
```

```
ggplot(data, aes(x = Height, y = Weight)) +
```

```
  geom_density_2d() +
```

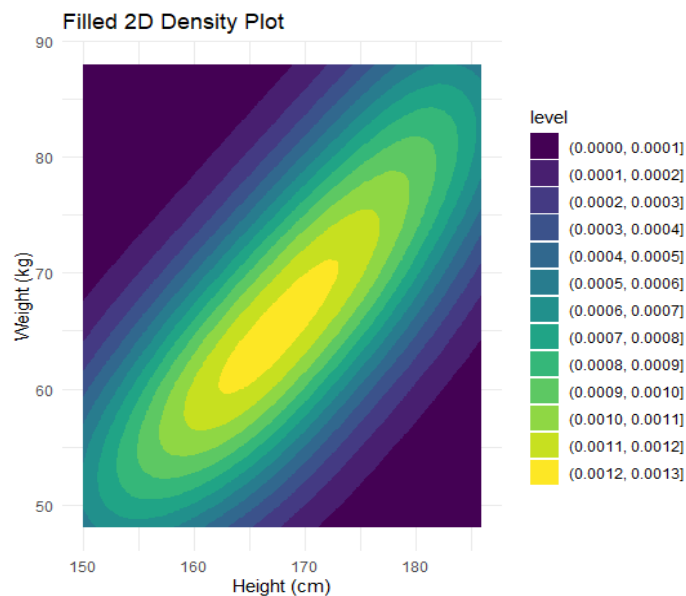
```
  geom_point() +
```

```
  labs(
```

```

    title = "2D Density Plot of Height and Weight",
    x = "Height (cm)",
    y = "Weight (kg)"
  ) +
  theme_minimal()
ggplot(data, aes(x = Height, y = Weight)) +
  geom_density_2d_filled() +
  labs(
    title = "Filled 2D Density Plot",
    x = "Height (cm)",
    y = "Weight (kg)"
  ) +
  theme_minimal()

```



Q5. Color Scales, Accessibility and Misleading Visualization

```

library(ggplot2)

Category <- c("A", "B", "C", "D", "E", "F", "G")
Satisfaction <- c(42, 48, 55, 61, 67, 74, 82)
data <- data.frame(Category, Satisfaction)

ggplot(data, aes(x = Category,

```

```

        y = Satisfaction,
        fill = Satisfaction)) +
geom_col() +
scale_fill_gradient2(
  low = "blue",
  mid = "white",
  high = "red",
  midpoint = 62
) +
labs(
  title = "Misleading Color Scale",
  x = "Category",
  y = "Satisfaction",
  fill = "Satisfaction"
) +
theme_minimal()
ggplot(data, aes(x = Category,
  y = Satisfaction,
  fill = Satisfaction)) +
geom_col() +
scale_fill_viridis_c(
  option = "viridis",
  direction = 1
) +
labs(
  title = "Satisfaction with Accessible Sequential Colors",
  x = "Category",
  y = "Satisfaction",
  fill = "Satisfaction"
) +

```

theme_minimal()

